

Project description and guidelines

IN 2022 Robotics, BSc Course, 2nd Sem., Dr. Julien Nembrini, Marta Visetti

Handout on Thursday April 30 2026

Week 8 to 13

Due on **Monday May 25 2026, midnight**

Presentation on **Thursday May 28 2026**

The remaining part of the semester will be devoted to a group project which consists of a *task* and a *challenge*. The challenge can be freely defined by the students but must be presented to the teaching team for approval before starting to work on it.

Important dates:

April 30 Final approval on challenge description

May 25, midnight Strict deadline for submitting the final project (incl. codes, report, and oral presentation draft)

May 28, 13h15 Final presentations

The techniques exercised during the first part of the semester are sufficient in combination to solve the task proposed. All behaviors needed are variants of Explorer/Lover behaviors, line and wall following or object detection controllers. However, it is allowed to use other techniques if wished. As during the first part of the semester, all robots should use the exact same code (unless especially agreed for the challenge).

Grade The final project is evaluated on the basis of:

- the task code/algorithm (25%)
- the challenge code/algorithm (25%)
- the report (30%)
- the video (10%)
- the final presentation (10%)

Important elements include:

- robustness and efficiency of the algorithm
- clarity and readability of the code
- structure and clarity of the report
- orthography and grammar

Hand in:

Upload your code, report (with links to a single video), and presentation draft (in PDF format) in a zip file on Moodle before the deadline mentioned above, (midnight). There will be no submission accepted beyond this deadline.

You may amend your presentation between the deadline and the final presentations, but in this case **you must resubmit it in PDF format per email before 12h00 on the day of the final presentations.**

Zip files not following the naming convention will not be considered.

Task

The task consists in making a robot map a simple environment, detect goal blocks relative to obstacle blocks and navigate between these blocks.

Environment

In the task, the robot arena is composed of a linear obstacle made with 3 or 4 white blocks and a pair of goal blocks (green and red) placed on either side of the obstacle.

The arena is fitted with an overhead camera that provides a top-down view in order to detect the ArUco marker fitted on top of the robot. Example code is provided to extract position and yaw angle.

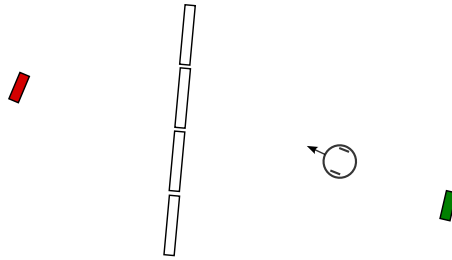


Figure 1: Task environment with obstacle blocks, goal blocks and initial position and orientation of the robot

At the beginning of the task, the mapping robot is positioned approximately at the origin of the tracking system facing the obstacle (Figure 1). The task goes through the following phases (in order)

Phase 1 Mapping: the robot explores the environment and builds a map of the arena, including the position of the blocks and the goal blocks.

Phase 2 Navigation: the robot uses the map to navigate between the goal blocks while avoiding the obstacle blocks. The robot should be able to reach the goal even if a different configuration is used.

Phase 3 Goal: the robot stops when reaching the second goal block.

For the task to be considered successful, the robot should follow the planned path, avoiding the obstacles.

Phase 1, “Mapping”

Task to achieve: construct a map of the environment, including the position of the blocks and the goal blocks.

1. the robot explores the environment and builds a map of the arena, including the position of the obstacle blocks and the goal blocks.
2. When reaching an obstacle block, the robot goes around it and then continues exploring.
3. the exploration may exploit the fact that the goal blocks are placed on either side of the obstacle blocks.

Phase 2, “Navigation”

Task to achieve: Efficiently and robustly use the map to navigate between one goal block to the other without colliding with the obstacle blocks.

1. Once reaching the second goal block, the algorithm should define a path to reach the first goal block detected during the mapping phase.
2. The robot should follow the path to reach the first goal block, while avoiding the obstacle blocks.
3. The navigation should be robust, meaning it should be able to handle unexpected situations and maintain performance under varying conditions.

Phase 3, “Goal”

Task to achieve: detect that the robot has reached the second goal block and stop.

1. the robot should stop when reaching the second goal block.
2. The goal detection should be robust, i.e. the robot should not take an obstacle block for a goal block, and should not miss the goal block when reaching it.
3. Upon reaching the goal, the robot lights up its leds.

Challenge

The challenge consists of defining, designing and solving an additional challenging task.

The challenge can be freely defined by the groups, but must be described in a short email (5 to 6 sentences at most). This idea will be discussed with the teaching team to get approval before starting work on it. A final discussion on project proposals will take place on **April 30**. Projects involving multiple robots are encouraged. Possible challenges are:

- use object recognition API
- neural network to improve robot behaviour
- communicate with other media (sound, light, etc)
- expand the use of the positioning system
- **your proposal**

Each group will have to choose a different challenge. If two groups choose something similar, variants that are sufficiently different will have to be proposed. Alternatively, multiple teams working on similar projects that involve competitive robots may test their controllers against other groups.

Guidelines

You are free to choose which strategy you implement for solving the task and the challenge, but keep the following guidelines in mind:

- You are expected to discuss the problem, including coding and testing, in group of two persons. Please inform the teaching team if you wish to change group before starting the project.
- **Report:** each student has to write her/his own report in Latex, maximum 12 pages + appendices (including this document as Appendix A). The report should present the project (task + challenge) as a whole. A link to an empty overleaf template is provided on moodle.
- **Code:** one code per group. We ask you to hand in a modular, readable and commented code. Your code will be thoroughly read.
- **Video:** one video per group, max 4 minutes, to display both task and challenge. An uninterrupted and accelerated shot of one race with both robots running your code must be included. Any acceleration must be annotated on the video and further annotations for clarity are encouraged. LED patterns should be visible.
- **Presentation:** each student has to do her/his own presentation (max. 4 minutes) focusing on a particularly interesting aspect of the project (task or challenge), different for each student. Introduction or conclusion need not be repeated. One presentation file per group is encouraged.

Task:

- The task relies on a good wall follower. Be careful to the robustness of the controller and the detection algorithm you may choose to use.
- Make your solution as robust (recover from error situations) and efficient as possible.
- Document your code (events, methods, algorithms, etc.), draw an FSM of your controllers.
- Use the LED patterns described in the appendix in order to indicate to an observer in which phases the robots are. Add the necessary custom patterns to explain your code, and document them using the same tabular presentation as in the appendix.

Appendix: LED patterns

The following is a list of the LED patterns that must be used. For each pattern, the indexes of the LEDs are given. The LEDs have to be turned on long enough to be visible in the video (use the plastic light diffusers for better LED visibility)

LED position	Corresponding action
Red LED 0/4	Waiting for position synchronization
RGB LED 3 colored	Color of the currently detected block
Red LEDs 2 or 6	Left or right wall followed
All Red LEDs (0/2/4/6) + body LED	Second goal reached
<i>Specific to your code</i>	Add to better explain your code